GUIDELINES FOR FLARE PAYMENTS

ISSUED BY

DEPARTMENT OF PETROLEUM RESOURCES

2020

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1. SCOPE

These Guidelines are issued pursuant to Flare Gas (Prevention of Waste and Pollution) Regulations, 2018, and shall apply to Flare Payments for natural gas being flared (flared or vented) at Producers' Production Facilities in Nigeria.

2. OBJECTIVE

The objective of these Guidelines is to define the accounting procedure for Flare Payments for the Producer at Production Facilities with respect to flared and/or Vented Gas.

3. FLARE PAYMENT REGIME

Flare Payments shall apply to any natural gas that is flared and/or vented at Production Facilities of the Producers.

Paragraph 12 of the Flare Gas (Prevention of Waste and Pollution) Regulations, 2018, stipulates that Permit Holders shall not engage in Routine Flaring or venting of natural gas and Paragraph 13 stipulates the applicable Flare Payment regime to the Producers.

4. FLARE PAYMENT AMOUNT ACCOUNTING PROCEDURES

4.1 Accounting Procedure during the Transition Period

a. During the Transition Period referred to in Section 4.9 of the Guidelines for Flare Gas Measurement Data Management and Reporting Obligations 2018, the Accounted Flare Gas quantity will be used as the basis for calculating the Flare Payments.

b. The Accounted Flare Gas quantities are calculated as follows:

Accounted Flare Gas quantities = (A* B) - C- D - E - F - G,

where:

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A = Associated Gas Production quantities (metered or accounted)

B = 1 - Shrinkage Factor

C = Own Consumption quantities (metered or accounted)

D = Gas for Existing Offtake Commitments (metered)

E = Liquids extracted from Associated Gas (metered or accounted)

F = Flare Gas delivered to Third Party Flare Commercialisation Projects (metered)

G = Flare Gas delivered for Producer's Approved Flare Out Project (metered).

c. The Chargeable Flare Gas Quantity equals the Accounted Flare Gas quantity less Contracted Guaranteed Flare Gas Buyer Shortfall.

d. The Flare Payment Amount is calculated by multiplying the Chargeable Flare Gas Quantity by the applicable Flare Payment rate in US$/Mscf.

Table 1: Producer's Flare Payment Amount Accounting Template (Transition Period)

A	B	C	D	E	F	G
OML/ Marginal Field ID	Crude oil Production in the OML/ Marginal Field (Bopd)	Accounted Flare Gas quantity (Mscf)	Contracted Guaranteed Flare Gas Buyer Shortfall (Mscf)	Chargeable Flare Gas quantity (Mscf) [C-D]	Applicable Flare Payment rate [B ≥ 10,000 bopd ($2.00/Mscf)] [B < 10,000 bopd ($0.50/Mscf)]	Flare Payment Amount due ($US) [G = E * F]

4.2 Accounting Procedure Post Transition Period

a. The Compounded Flare Gas Quantities is the sum of Flare Gas quantities and Unaccounted Flare Gas quantities.

b. The Chargeable Flare Gas Quantity equals the Compounded Flare Gas Quantity less the Contracted Guaranteed Flare Gas Buyer Shortfall.

c. The Flare Payment Amount is calculated by multiplying the Chargeable Flare Gas Quantity by the Applicable Flare Payment in US$/Mscf.

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Table 2: Producer's Flare Payment Amount Accounting Template (Post Transition Period)

A	B	C	D	E	F	G	H
OML / Marginal Field ID	Crude Oil Production in the OML / Marginal Field (Bopd)	Flare Gas quantity - metered (Mscf)	Unaccounted Flare Gas quantities (Mscf)	Contracted Guaranteed Flare Gas Buyer Shortfall (Mscf)	Chargeable Flare Gas quantity (Mscf) [C+D-E]	Applicable Flare Payment rate [B ≥ 10,000 bopd ($2.00/Mscf)], [B < 10,000 bopd ($0.50/Mscf)]	Flare Payment Amount ($US) [H = F * G]

5. FLARE PAYMENT ADMINISTRATION

5.1 Flare Payment Procedure

a. The Department of Petroleum Resources shall collate and reconcile reported monthly oil production figures in OMLs and Marginal fields.

b. Producers must present the data in accordance with the Guidelines for Flare Gas Measurement Data Management and Reporting Obligations.

c. An average monthly oil production, taking into consideration only days during which oil was produced, will be used to establish the applicable Flare Payment.

d. The various elements that contribute to the determination of the Chargeable Flare Gas Quantity shall be validated by the Department of Petroleum Resources.

e. The Flare Payment Amount shall be computed monthly as a product of the Chargeable Flare Gas Quantity in thousand standard cubic feet (Mscf) multiplied by the applicable Flare Payment rate.

f. Flare Payment Amounts shall be paid by the Producer in accordance with the procedures for payment of royalties to the Federal Government of Nigeria, provided for in Section 3 of the Associated Gas Re-Injection Act 1979.

g. Reporting shall be done in accordance with Paragraph 19 of the Flare Gas (Prevention of Waste and Pollution) Regulations 2018.

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5.2 Flare Payment Additional Considerations

a. No Flare Payment Amounts shall become due in respect of an agreed volume of Flare Gas that the Producer is committed to deliver to the Permit Holder under a Deliver or Pay Agreement entered into with the Permit Holder as from the commencement of commercial operations of the Project, as stipulated in Paragraph 13 (3) of the Regulations.

b. The provision under Section 5.2 (a) above ceases upon expiration of the term of the Gas Sales Agreement signed with the Permit Holder for the applicable Project.

6. NON-COMPLIANCE AND PENALTY

6.1 Non-Compliance

The additional payment, referred to in Paragraph 21 of the Flare Gas (Prevention of Waste and Pollution) Regulations 2018, applies where a Producer fails to meet the obligations as stated in Paragraph 21 (1) of the Flare Gas (Prevention of Waste and Pollution) Regulations, 2018. This includes where a Producer inhibits or disrupts the process of the Federal Government of Nigeria in granting Permits to Access Flare Gas or delivering natural gas under the Gas Sales Agreement as stipulated in Paragraph 21 (1) (c) and (d).

6.2 Late Payment

Failure to remit the Flare Payment Amount by the Producer shall attract applicable sanctions as provided for in Section 40 of the First Schedule of the Petroleum Act 1969.

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7. GLOSSARY

Accounted Flare Gas	The Accounted Flare Gas quantities are calculated as follows: Accounted Flare Gas quantities = (A* B) - C- D - E - F – G, where A= Associated Gas Production quantities (metered or accounted), B = Shrinkage Factor, C = Own Consumption quantities (metered or accounted), D = Gas for Existing Offtake Commitments (metered), E = Liquids extracted from Associated Gas (if metered or accounted for), F = Flare Gas delivered to Third Party Flare Commercialisation Projects (metered), and G = Flare Gas delivered for Producer`s Approved Flare Out Projects (metered). During the Transition Period the Accounted Flare Gas quantity will be used as the basis for calculating the Flare Payments.
Associated Gas (AG)	Gas associated with Crude Oil or Condensate production
Associated Gas Production	Quantities of Associated Gas produced
Buyer Gas Connection Assets	The natural gas pipeline used to transport Flare Gas from the Delivery Point at the perimeter of the Flare Site to the Project Facility and any other equipment, machinery or other property of any kind that are owned or leased by the Flare Gas Buyer to take delivery of Flare Gas under the Connection Agreement
Chargeable Flare Gas Quantity	The Flare Gas quantity used for calculating the Flare Payment to be paid by the Producer as defined in the Guideline for Flare Payments

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Compounded Flare Gas Quantities	The sum of the Flare Gas quantities and Unaccounted Flare Gas quantities
Condensate	A low-density, high-API gravity liquid hydrocarbon phase that generally occurs in association with natural gas
Contracted Guaranteed Flare Gas	The Guaranteed Flare Gas quantity contracted by the Flare Gas Buyer under the Gas Sales Agreement, and for which Flare Gas Buyer has entered into a Deliver or Pay Agreement with the Producer
Contracted Guaranteed Flare Gas Buyer Shortfall	Contracted Guaranteed Flare Gas that was made available by the Producer under the Deliver or Pay Agreement which the Flare Gas Buyer did not take
Crude Oil	Oil in its natural state before it has been refined or treated (excluding water and other foreign substances)
Deliver or Pay Agreement	An agreement that the Producer may sign with the Flare Gas Buyer under which the Producer guarantees to supply an agreed quantity of gas to the Flare Gas Buyer
Delivery Point	Point within the Metering Station at which title of the Flare Gas passes from the Seller to the Flare Gas Buyer
Department of Petroleum Resources (DPR)	The Department of Petroleum Resources of the Federal Ministry of Petroleum Resources, or any successor entity that has statutory responsibility for ensuring compliance with petroleum laws, regulations and guidelines related to the oil and gas industry in Nigeria
Existing Offtake Commitment	Gas delivery commitment that the Producer has in place for the delivery of Associated Gas to a third party or a Producer Entity prior to the Regulations becoming effective

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Flare Gas	Any natural gas produced in association with crude oil by a Producer and finally diverted toward a Flare Site by the Producer with the intent that the natural gas will be flared, including any such natural gas from a Greenfield Project
Flare Gas Buyer	Preferred Bidder which has, as Flare Gas Buyer, executed the Final Commercial Agreements
Flare Gas Connection Point	The point where the Gas Connection Assets tie into the facilities of the Producer
Flare Gas Flared	Flare Gas volumes flared at a Flare Stack
Flare Gas Header	The gas facility or facilities within the Producer`s existing infrastructure from where Associated Gas is directed to the Flare Stack (Flare Gas Connection Point may or may not be at the Flare Gas Header)
Flare Gas (Prevention of Waste and Pollution) Regulations 2018	The Regulations by this name that were signed into effect on the 5th of July 2018, also referred to as the Regulations in the Guidelines
Flare Gas-to-Market Product	A product delivered by the Project to an Off-Taker after having been processed and/or transported by the Flare Gas-to-Market Technology
Flare Payment	Payment made by the Producer in US Dollars for each Mscf of Flare Gas flared or vented as defined in the Regulations
Flare Payment Amount	The payment made by the Producer to the Federation Account of the Federal Government of Nigeria as stipulated in the Flare Gas (Prevention of Waste and Pollution) Regulation 2018 based on the Flare Payment and the Chargeable Flare Gas quantity

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Flare Site	As defined in the Regulations, a location where natural gas produced in association with Crude Oil and/or Condensate is flared, commencing at a Flare Gas Header and going to the point of the flare within an Oil Mining Lease or Marginal Field area or within an oil terminal or refinery. A Producer may consolidate operations for more than one production area into one or more Flare Sites. Flare Stacks connected to different Flare Gas Headers are considered to be separate Flare Sites.
Flare Stack	Gas combustion device used at the Flare Site or at the Project to burn Flare Gas
Gas Connection Assets	Assets that include: (a) Buyer Gas Connection Assets; and (b) Producer Gas Connection Assets
Gas Sales Agreement	An agreement, conforming substantially to the template appended to the request for proposal by the Federal government of Nigeria in the Programme, whereby Flare Gas is sold by the Seller to the Flare Gas Buyer
Greenfield Project	Any project to develop new oil or natural gas production from an Oil Mining Lease or Marginal Field after the effective date of the Flare Gas (Prevention of Pollution and Waste) Regulations, 2018
Marginal Field	As defined in the Regulations, an Oil Field within an Oil Mining Lease area awarded to an applicant in relation to a field identified as a marginal field by the President of the Federal Republic of Nigeria
Metering Station	A gas measuring, control and registering station funded, designed, equipped and built according to industry standards by the Flare Gas Buyer. This station incorporates the Delivery Point and is located

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	within the Producer's property at the perimeter of the Flare Site and forms part of the Producer Connection Assets. This station should register, amongst others, the following parameters: • Pressure • Temperature • Flow rate
Minister	The Minister of Petroleum Resources
Non-Associated Gas	Free natural gas not in contact with, or dissolved in, Crude Oil in the reservoir. There may or may not be condensate production together with the gas.
Non-Routine Flaring	As defined in the Regulations, all flaring other than Routine Flaring. Non-routine flaring is typically intermittent and of short duration and either planned or unplanned. This includes: I. Subject to prudent operations and prompt action on the side of the Producer, temporary (partial) failure of equipment that handles the gas during normal operations until their repair or replacement, e.g. failure of compressors, pipeline, instrumentation, controls; II. Temporary failure of facilities that prevents receipt of Associated Gas; III. Safety Flaring; IV. Initial plant or field start-up before the process achieves steady operating conditions; V. Start-up following facility shutdowns; VI. Scheduled preventive maintenance and inspections subject to it being carried out in a prompt manner and following prudent

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	industry standards; VII. Construction activities, such as tie-ins, abrupt change of operating conditions, plant design modifications; VIII. Process upsets when process parameters fall outside the allowable operating or design limits and flaring is required to stabilise the process; IX. Reservoir or well maintenance activities such as acidification, wire line interventions; X. Exploration, appraisal, or production-well testing or clean-up following drilling or well work-over
Oil Field	A geographic area under which an oil reservoir lies, and/or a tract of land used for extracting petroleum, otherwise known as Crude Oil, from the ground
Oil Mining Lease (OML)	A lease granted by the Minister to a company incorporated in Nigeria, which allows such company to search for, win, work, carry away, and dispose of oil
Own Consumption	Associated Gas quantity consumed by the Producer in the process of production of Crude Oil and/or Condensate (for power generation, heating, artificial lift, re-injection, etc.)
Permit Holder	A company that has, pursuant to the Regulations, been granted a Permit to Access Flare Gas by the Minister
Permit to Access Flare Gas	As defined in the Regulations, a permit granted to a Permit Holder by the Minister to take Flare Gas on an exclusive basis from one or more Flare Sites as designated in the permit on behalf of the Federal Government of Nigeria.
Preferred Bidder	The status given to a Bidder whose proposal for one or more Flare

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	Site(s) is one of the proposals that make up the highest-ranking proposals
Processing Facility	Production Facility or any other facility where natural gas or Crude Oil is processed. Power plants are not considered to be processing facilities.
Producer	A holder of Oil Mining Lease or allottee of a Marginal Field or a contractor under a Production Sharing Contract
Producer Entity	A midstream entity incorporated in Nigeria that is not a Producer which, following the provisions of the Flare Gas (Prevention of Waste and Pollution) Regulations, 2018, and the Guidelines issued thereunder, can become a Flare Gas Buyer
Producer Gas Connection Assets	The pipeline, equipment, machinery, other assets or facilities including the Measuring Station designed, funded and built by the Permit Holder and used to transport Flare Gas from the Flare Gas Connection Point to the Delivery Point under the Connection Agreement. Title, care and custody of these assets are transferred by the Permit Holder to the Producer on the Commercial Operations Date.
Producer's Approved Flare Out Project (PAFOP)	As defined in the Regulations, a Flare Gas Commercialisation project intended to be developed by a Producer that is not already operating commercially and that meets the following criteria as validated and approved by the Department of Petroleum Resources: i. received all required permits / licenses necessary for its operation; ii. executed all material commercial agreements necessary for

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	the implementation of the project; iii. achieved financial closing; iv. has demonstrated the viability of achieving commercial operations by no later than 1 January 2020; and v. the Producer has provided a financial guarantee to the Federal Government of Nigeria to underpin its commitment to milestones for the implementation of the Project
Production Facility	Facility where the Producer processes fluids separating and handling Crude Oil, Condensate, water and Associated Gas
Programme	The Nigerian Gas Flare Commercialisation Programme or any other program to take Flare Gas at any Flare Site on behalf of the Federal Government of Nigeria through a competitive bid process and/or through authorisation of Producers' Approved Flare Out Projects
Project	A technically and commercially viable and sustainable Flare Gas utilisation project developed by the Flare Gas Buyer either as a Third Party Flare Commercialisation Project or a Producer's Approved Flare Out Project
Project Facility	The facilities used in the Project, including the surrounding facilities and property, at which a Flare Gas Buyer takes Flare Gas as fuel and/or feedstock and which produce and/or transport a Flare Gas-to-Market Product
Project Flare Gas Flared	Flare Gas flared at the Producer's Flare Stack attributable to Contracted Guaranteed Flare Gas Buyer Shortfall and/or vented or flared at the Flare Stack at a Project
Regulations	Flare Gas (Prevention of Waste and Pollution) Regulations, 2018
Routine Flaring	Flaring of natural gas produced in association with Crude Oil and/or

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	Condensate during normal oil production operations in the absence of sufficient facilities or amenable geology to re-inject the produced gas, utilise it on site, or dispatch it to a market. Routine Flaring does not include safety flaring even when continuous.
Seller	The Federal Government of Nigeria, or an entity that is assigned or designed by the Federal Government of Nigeria with respect to the Gas Sales Agreement, and that shall serve as the Seller of the Flare Gas to a Flare Gas Buyer
Shrinkage Factor	0.30%, which is the estimated percentage of the Associated Gas Production quantity that, due to pressure and temperature variations prior to any metering downstream of the Crude Oil Separators, will transform from gas to liquid phase
Third Party Flare Gas Commercialisation Project	A Project awarded under any competitive bid process under the Programme conducted by the Federal Government of Nigeria to take Flare Gas on its behalf any Flare Site as set forth in the Regulations
Transition Period	Period during which the Producers and Processing Facility operators must install the metering system as laid out in the Guidelines for Flare Gas Measurement Data Management and Reporting Obligations
Unaccounted Flare Gas	Flare Gas quantities that Producers cannot account for, calculated as (A*B) - C - D - E - F - G – H, where: A = Associated Gas Production quantities B = 1 - Shrinkage Factor C = Own Consumption quantities D = gas quantities delivered under Existing Offtake Commitments

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	E = Liquids extracted from Associated Gas (if metered or accounted for) F = Flare Gas delivered to Third Party Flare Commercialisation Project G = Flare Gas delivered for Producers Approved Flare Out Project H = measured or accounted Flare Gas quantities going to the Flare Stack
Vented Gas	Gas that is released into the environment without being combusted
Vented Gas Quantity	The quantity of Flare Gas vented either by a Producer, a Permit Holder or a Processing Facility

Approved by		
Engr. Sarki Auwalu, ***MNSE*** **(Director/CEO, Department of Petroleum Resources)**		
	Date	**1st August 2020**

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